

Steven Caro

Waterloo, Ontario

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🌐 [Website](#)

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SUMMARY

Self-motivated, disciplined, and eager to learn new skills. Excited to apply my knowledge of Machine Learning to real-world problems. I have a strong background in Electrical Engineering and Computer Science, with experience in deep learning, software development, controls engineering, and power distribution analysis. I am passionate about robotics and automation, and I am looking for opportunities to contribute to innovative projects in these fields.

EXPERIENCE

Graduate Researcher

Autonomous Systems Lab, Vector Institute & University of Waterloo 01/2024 - 12/2025
Waterloo, Ontario

- Optimize policies for autonomous robots using PyTorch and state-of-the-art Machine Learning techniques including deep reinforcement learning and generative diffusion models
- Coupling ML with Computer Vision to develop efficient manipulation policies for robotics

Software Developer

Human Systems Lab, University of Windsor 05/2020 - 08/2023
Windsor, Ontario

- Used Python to create diagnostic tests used in conjunction with an eye-tracking device
- Used the data accumulated from the eye-tracking device to generate a report that optometrists use to diagnose common optical diseases

Electrical Engineering Intern

Hiram Walker & Sons Ltd 09/2022 - 12/2022
Windsor, Ontario

- Completed electrical designs for fire safety system
- Programmed PLCs using RSLogix 5000
- Assisted in programming logic for quality assurance on bottle lines and defect detection using x-ray & computer vision
- Programmed HMIs using FactoryTalk View Studio ME/Site
- Used AutoCAD for electrical drawing updates

Controls Engineering Co-op

Onyx Engineering Inc 01/2022 - 04/2022
Windsor, Ontario

- Implemented circuit design modifications using AutoCAD Electrical
- Trained in SCADA & HMI Industrial Automation
- Used Visual Basic, macros, and pivot tables to import and analyze daily website traffic data

Power Distribution Engineering Analyst

Entegrus Powerlines Inc 05/2021 - 08/2021
Chatham, Ontario

- Performed load flow and voltage drop studies to determine acceptable transformer and conductor sizes for new projects
- Assisted in creating and balancing the 2022 budget by performing cost-estimation on proposed projects, and prioritizing the projects by metrics such as customer impact, safety, and reliability

EDUCATION

University of Waterloo 01/2024 - 12/2025
MASC - Electrical and Computer Engineering
(GPA: 3.9/4.0)

University of Windsor 09/2019 - 08/2023
BASC Honours Electrical and Computer Engineering
(GPA: 4.0/4.0)

SKILLS

Python PyTorch Excel MATLAB
MicroPython Raspberry Pi Fusion 360
3D Printing RSLogix SCADA
AutoCAD Electrical LTSpice

PROJECTS

Created a Biomimetic Robotic Hand
01/2023 - 08/2023 Engineering Capstone project in which a wearable robotic hand mimics the actions of the user's hand

Team Leader - Singam Labs Coding Club
05/2021 - 08/2021 Led group of classmates in a python coding club by walking through my visual cognitive-testing software to demonstrate different programming techniques

Constructed a full-size Arcade Machine
06/2020 - 09/2020 Complete bottoms-up build, including cabinetry and electronics

AWARDS/FUNDING

NSERC CGSM Research Grant: Valued at \$27,000.

Engineering Excellence Master's Fellowship: Valued at \$25,000/year. (University of Waterloo)

President's Entrance Award: Valued at \$4000/year. (University of Windsor)

Outstanding Scholar Award: Funding for undergraduate research valued at \$3000/year. (University of Windsor)